

Unit 10 Notes: Measures of Spread

Measures of spread are used to describe the distribution, or spread, of the data. The range is the difference between the greatest and least data values. Quartiles are values that divide the data set into four equal parts. The median of the lower half of a set of data is the 1st quartile and the median of the upper half of a set of data is the 3rd quartile. The difference between the third quartile and the first quartile is called the Inter quartile range (IQR).

Example 1

Determine the measures of spread for the number of votes received for student government president:

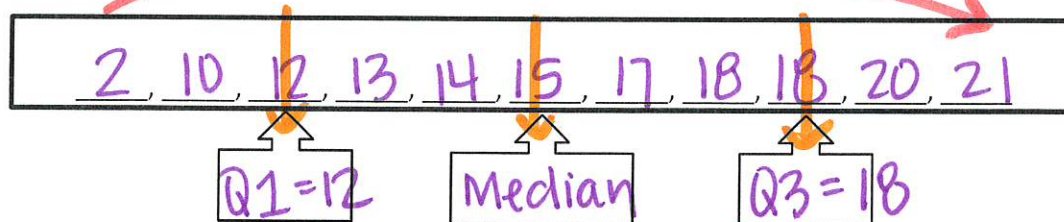
The greatest number in the data set is 21.

The least number is 2.

13, 20, 18, 12, 21, 2, 18, 17, 15, 10, & 14 votes.

The range is 21 - 2 or 19 votes.

To determine the quartiles, arrange the numbers in order from least to greatest:



The interquartile range is 18 - 12 or 6.

An outlier is a data value that is either much greater or much less than the median.

Determine the range, median, first and third quartiles and interquartile range for each data set.

Number of Boxes of Popcorn Sold						
52	72	96	21	58	40	75

21, 40, 52, 58, 72, 75, 96



Range: 96 - 21 = 75

Median: 58

Quartile 1: 40

Quartile 3: 75

IQR: 75 - 40 = 35

Test Grades									
83	83	85	87	89	88	87	79	81	
67	79	81	83	83	85	87	88	89	
Q1 = 80			Median			Q3 = 87.5			

Number of Text Messages Sent							
20	23	18	4	17	21	15	56
4	15	17	18	20	21	23	56
Q1 = 16		Median		Q3 = 22			

Range: $89 - 67 = 22$

Median: 83

Quartile 1: 80 $\frac{(79+81)}{2}$

Quartile 3: 87.5 $\frac{(87+88)}{2}$

IQR: $87.5 - 80 = 7.5$

Range: $56 - 20 = 36$

Median: 19 $\frac{(18+20)}{2}$

Quartile 1: 16

Quartile 3: 22

IQR: $22 - 16 = 6$

Raul and Bryan recorded the number of hours they slept last week. The data is shown below.

Raul's Sleep Log						
8	8.5	9.5	9	8.5	7	8.5

7, 8, 8.5, 8.5, 8.5, 9, 9.5

Q1 = 8, M = 8.5, Q3 = 9

IQR: $9 - 8 = 1$

Bryan's Sleep Log						
6.5	7	7.5	6.5	12	7	7.5

6.5, 6.5, 7, 7, 7.5, 7.5, 12

Q1 = 6.5, M = 7, Q3 = 7.5

IQR: $7.5 - 6.5 = 1$

1. Which statement is NOT true about the data provided?

A. Both sets of data have an interquartile range of 1. ✓

B. Both measures of center for Raul's data have a value of 8.5. False

C. The shapes of both data distributions are non-symmetric. ✓

D. The IQR of Bryan's data is best used to describe the spread because of the outlier. ✓

2. Which is a true comparison of the measures of center for Bryan's data?

A. Range = Mean $1 \neq 7.7$

B. Mean < Median $7.7 < 7$

C. Median > IQR $7 > 1$ True

D. Mean < Median

$$\begin{array}{r} 2 \\ 15 \\ 17 \\ 9 \\ +18 \\ \hline 59 \end{array}$$

$$\begin{array}{r} 8.4 \\ 7 \overline{)59.00} \\ \underline{-56} \\ 30 \\ \underline{-28} \\ 20 \end{array}$$

$$\begin{array}{r} 13 \\ 14 \\ 15 \\ +12 \\ \hline 54 \end{array}$$

$$\begin{array}{r} 7.71... \\ 7 \overline{)54.00} \\ \underline{-49} \\ 50 \\ \underline{-49} \\ 10 \\ \underline{-7} \\ 3 \end{array}$$